

Menglin Zhu

Curriculum Vitae with References

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Department of Materials Science and Engineering
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Research Interest

- Probing quantum/energy materials atom by atom using advanced (S)TEM:
 - Multislice electron ptychography ▪ *Operando* microscopy ▪ 4D-STEM ▪ Spectroscopy
- Modeling atomistic behavior and phase transitions using DFT and *ab initio* MD simulations.
- Extracting insights from microscopy data using statistical analysis and machine learning

Education and Training

- 2023–present **Postdoctoral Researcher**, *Massachusetts Institute of Technology*, Prof. James M. LeBeau.
2018–2023 **Ph.D. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.
2014–2017 **B.S. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.
2013–2014 **Materials Science and Engineering**, *Nanjing University of Aeronautics and Astronautics*.

Honors & Awards

- 2026 **Albert Crewe Award** of Microscopy Society of America to recognize distinguished contributions to the field of microscopy in the physical sciences of an early career scientist
- 2025 **PARADIM user proposal #366** *Probing Dynamic Polar Structures in Relaxors*, 10 days of microscope time at Cornell PARADIM center
- 2023 **Postdoctoral Scholar** of Microscopy Society of America, awarded as a postdoctoral researcher for the paper submitted for the Microscopy and Microanalysis conference 2023
- 2022 **Student Scholar** of Microscopy Society of America, awarded as a Ph.D. candidate for the paper submitted for the Microscopy and Microanalysis conference 2022
- 2021 **Presidential Fellow** of Ohio State University, the most prestigious award given by the Graduate School to outstanding Ph.D. candidates
- 2017 **The Mars Fontana Scholarship** by Ohio State University, awarded to the most outstanding senior student in the Department of Material Science and Engineering
- 2016 **Summer Research Fellow** of Ohio State University
Markworth-Woolley Scholarship by Ohio State University

Publications

Citations: 1439, h-Index: 17, *Equal Contribution, #Correspondance

Teaching Experience

- Fall, 2023 **3.34 Imaging of Materials**, guest lecturer, Massachusetts Institute of Technology
- Fall, 2020 **MSE3151 Transport Phenomonon and Kinetics**, teaching assistant, Ohio State University
- Fall, 2019 **MSE3332 Undergraduate Lab II**, laboratory assistant and instructor, Ohio State University

Synergistic Activities

- 2023–present **Mentor of two graduate students** on project *Collaborative for Hierarchical Agile and Responsive Materials (CHARM) under cooperative agreement W911NF-19-2-011*

- 2023 **Co-organize Microscopy and Microanalysis conference P08 symposium** entitled *Atomic Scale Microscopy of Interfaces and Heterostructures with Correlated Phenomena*
- 2023 **Mentor of two REU Students** on project *NFO Thin Films Grown via an Off-Axis Sputtering Method; STEM Characterization of NdFeO₃/SrTiO₃ Thin Films*
- 2022 **Mentor of one REU Student** on project *Effects of Octahedral Tilting and Lattice Strain on LaFeO₃*

References

Advisor **Prof. Jinwoo Hwang**, Associate Professor
Department of Materials Science and Engineering, the Ohio State University
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Advisor **Prof. James M. LeBeau**, Associate Professor
Department of Materials Science and Engineering, Massachusetts Institute of Technology
Email: lebeau@mit.edu

Collaborator **Prof. Lane W. Martin**, Robert A. Welch Professor
Department of Materials Science and NanoEngineering, Chemistry, and Physics and Astronomy,
Rice University
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Collaborator **Prof. Xiaolei Guo**, Assistant Professor
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